# 7.1 An Introduction to A Level Organic Chemistry (A Level Only)

## Question Paper

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| Course | CIE A Level Chemistry |
| Section | 7. Organic Chemistry (A Level Only) |
| Topic | 7.1 An Introduction to A Level Organic Chemistry (A Level Only) |
| Difficulty | Hard |

**Time allowed:** 40

**Score:** /32

**Percentage:** /100

## Question 1a

This question is about aldehydes.

Draw the displayed formula and 3D representations of the smallest aldehyde that can form optical isomers.

| displayed formula |
|---|
| |

| optical isomer 1 | optical isomer 2 |
|---|---|
| | |

**[3 marks]**

## Question 1b

i)

Draw the fully displayed formula of the oxidation product of the aldehyde identified in part **(a)**.

[1]

ii)

State a suitable reagent, including observations, for the oxidation of the aldehyde identified in part **(a)**.

[1]

**[2 marks]**

## Question 1c

i)

Write a balanced symbol equation, using structural formulae, to show the reduction of the aldehyde identified in part **(a)**. You should use [H] to show the reducing agent.

[1]

ii)

State a suitable reducing agent.

[1]

**[2 marks]**

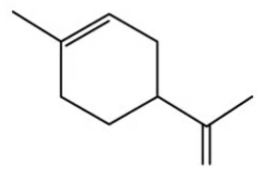
## Question 1d

Name the organic compound formed by the reaction of the oxidation product in part **(b)** and the reduction product in part **(c)**.

**[1 mark]**

## Question 2a

This question is about the chemistry of limonene.



Limonene is commonly found in the rinds of citrus fruits such as grapefruit, lemon, lime and oranges. It is a naturally occurring hydrocarbon with the molecular formula $C_{10}H_{16}$.

Limonene exists as a pair of enantiomers:

- One enantiomer is responsible for a strong orange smell
- The other enantiomer is responsible for a lemon smell.

Draw 3D representations of the **two** enantiomers of limonene.

| optical isomer 1 | optical isomer 2 |
|---|---|
| | |

**[2 marks]**

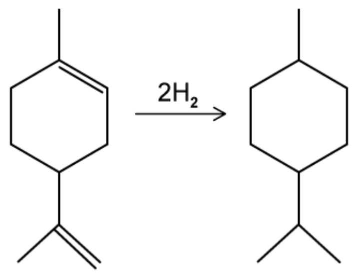
## Question 2b

Suggest why receptors in the human nose can distinguish between the orange and lemon enantiomers of limonene.

**[3 marks]**

## Question 2c

Limonene can undergo full hydrogenation to form menthane as shown.




For the complete hydrogenation of an impure sample of 3.00 g of limonene at room temperature and pressure, 870 $cm^3$ of hydrogen was required.

Calculate the percentage purity of limonene.

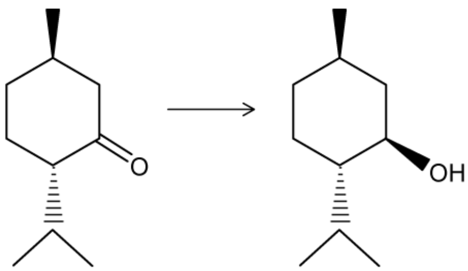
(1 mol of hydrogen occupies 24.0 $dm^3$ at room temperature and pressure)

percentage purity = ........................ %

**[4 marks]**

## Question 2d

The final step of the multi-step synthetic conversion of limonene to menthol is shown.




State the reagent for this reaction and the type of reaction.

**[2 marks]**

## Question 3a

Aldehydes and ketones always react with HCN to form optical isomers.

Discuss this statement, using the general formulae RCHO and R'COR''. Your answer should include equations where appropriate.

**[4 marks]**

## Question 3b

Acidified $K_2CrO_7$ (aq) was heated under reflux with the following β-hydroxyaldehyde shown in Fig. 3.1.

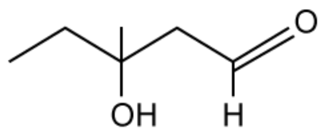


**Fig. 3.1**

Explain the change in chirality, if any, between the β-hydroxyaldehyde and its organic product.

**[3 marks]**

## Question 3c

Explain the change in chirality, if any, when pent-1-en-3-ol is hydrogenated.

**[3 marks]**

## Question 3d

Other isomers of pent-1-en-3-ol, $C_5H_{10}O$, exist.

i)

Draw the displayed formula of the secondary alcohol isomer of pent-1-en-3-ol that exhibits both stereoisomerism and optical isomerism and identify the chiral centre.

[2]

ii)

Give the IUPAC name for this isomer.

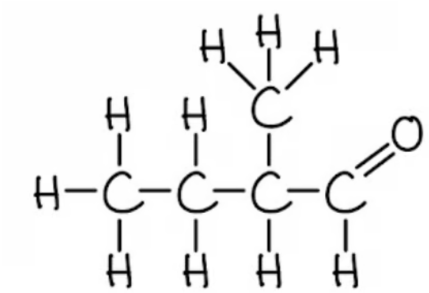
[1]

**[3 marks]**

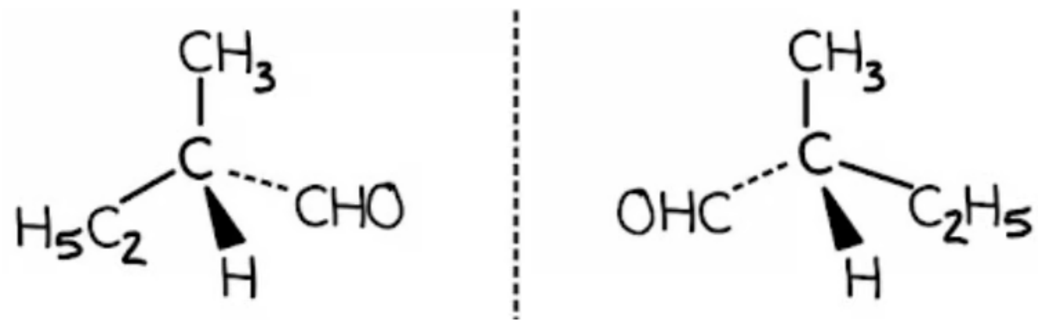
# Model Answers: Hard

1a

a) The displayed formula and 3D representations of the smallest aldehyde that can form optical isomers are:



- Correct displayed formula; [1 mark]



- Correct 3-dimensional structure of the left-hand isomer; [1 mark]
- Correct 3-dimensional structure of the right-hand isomer; [1 mark]

**[Total: 3 marks]**

- You need a carbon with four different atoms or groups of atoms to form optical isomers
  - Therefore, the aldehyde carbon **cannot** be the chiral carbon
- The chiral carbon must have an aldehyde group attached, which leaves 3 bonds for other atoms or groups of atoms
- To create the smallest aldehyde with optical isomer, you should add a hydrogen, methyl group and ethyl group to the chiral carbon so that you are adding

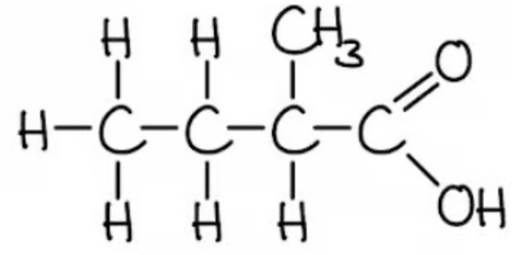
different hydrocarbon functional groups

- **Careful:** Do not add other functional groups because the question asks for the smallest aldehyde

1b

b)

i) The displayed formula of the oxidation product of the aldehyde identified in part **(a)** is:

-  ; [1 mark]

ii) A suitable reagent, including observations, for the oxidation of the aldehyde identified in part **(a)** is:

**Alternative 1**

- Reagent = Fehling's solution
  **AND**
  Observation = blue solution to brick red precipitate; [1 mark]

**Alternative 2**

- Reagent = Tollens' reagent
  **AND**
  Observation = formation of a silver mirror; [1 mark]

**Alternative 3**

- Reagent = Acidified potassium dichromate(VI) solution under reflux
  **AND**
  Observation = orange solution to green solution; [1 mark]

**[Total: 2 marks]**

- **Remember:** Aldehydes oxidise to carboxylic acids
- The wording of this question can be a bit off-putting but it is basically asking "what is formed when you oxidise an aldehyde?"
- You are expected to know this along with the reagents and observations as one of the more basic points of organic chemistry
- Although reagents and their associated observations are typically for 2 marks, you may sometimes see them for one combined mark where you have to get the combined reagent and observation matching and correct

1c

c)

i) The balanced symbol equation to show the reduction of the aldehyde identified in part **(a)** is:

- $C_2H_5CH(CH_3)CHO + 2[H] \rightarrow C_2H_5CH(CH_3)CH_2OH$; [1 mark]

ii) A suitable reagent is:

- Sodium tetrahydridoborate(III) / sodium borohydride / $NaBH_4$
  **OR**
  Lithium tetrahydridoaluminate(III) / lithium aluminium hydride / $LiAlH_4$; [1 mark]

**[Total: 2 marks]**

- **Remember:** Aldehydes reduce to primary alcohols
- You are expected to know this along with the reagents required for this reduction as part of your organic chemistry as well as being able to apply your equation writing skills to questions such as these

1d

d) The organic compound formed by the reaction of the oxidation product in part **(b)** and the reduction product in part **(c)**:

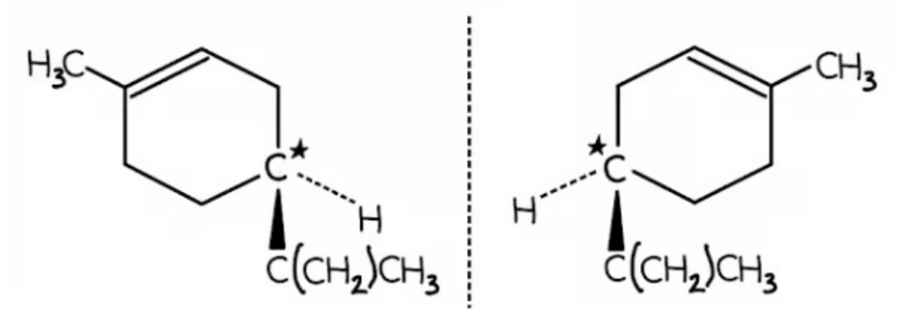
- 2–methylbutyl-2–methylbutanoate; [1 mark]

**[Total: 1 mark]**

- In part (b), you drew the structure of 2-methylbutanoic acid
- In part (c), you write the structural formula of 2-methylbutan-1-ol
- If you think about the reaction of ethanol and ethanoic acid:
  - Ethanol + ethanoic acid → ethyl ethanoate + water
- Then apply this principle but substituting the names for the alcohol and acid in this question
  - 2-methylbutanol + 2-methylbutanoic acid → **2-methylbutyl-2-methylbutanoate** + water

2a

a) The 3D representations of the two optical isomers produced are:




- Correct 3-dimensional structure of the left-hand isomer; [1 mark]
- Correct 3-dimensional structure of the right-hand isomer; [1 mark]

**[Total: 2 marks]**

- When you rule out the carbon atoms that have more than one hydrogen attached or are part of a C=C bond, you are left with one carbon which must be the chiral centre / asymmetrical carbon
- The key to **easily** drawing the 3D representations of the optical isomers for this compound is to keep the ring structure as the two flat bonds / in the plane
- There are 12 other possible structures depending on the bonds that you use for the ring structure and where you position the hydrogen and the alkenyl chain but these are generally more complicated to draw

2b

b) Receptors in the human nose can distinguish between the orange and lemon enantiomers of limonene because:

- Each enantiomer / optical isomer has a specific 3D structure; [1 mark]
- Receptors in the human nose have their own specific 3D structure
  **OR**
  Receptors in the human nose are stereospecific; [1 mark]
- Therefore, only one enantiomer fits in a certain receptor
  **OR**
  Therefore, one set of receptors can only smell orange and another set of receptors can only smell lemon; [1 mark]

**[Total: 3 marks]**

- The theory behind your ability to distinguish between the smell of oranges and lemons is based on the concept in biology of how molecules fit together and interact and that you appreciate the specific shapes of molecules
- Tells you that you are dealing with enantiomers
- You should know that enantiomers are non-superimposable mirror images of one another

enantiomers of limonene because:

- Each enantiomer / optical isomer has a specific 3D structure; [1 mark]
- Receptors in the human nose have their own specific 3D structure
  **OR**
  Receptors in the human nose are stereospecific; [1 mark]
- Therefore, only one enantiomer fits in a certain receptor
  **OR**
  Therefore, one set of receptors can only smell orange and another set of receptors can only smell lemon; [1 mark]

**[Total: 3 marks]**

- The theory behind your ability to distinguish between the smell of oranges and lemons is based on the concept in biology of how molecules fit together and interact and that you appreciate the specific shapes of molecules
- Tells you that you are dealing with enantiomers
- You should know that enantiomers are non-superimposable mirror images of one another
- Therefore, enantiomer 1 might fit into receptor A but enantiomer 2 cannot
- At the same time, enantiomer 2 might fit into receptor B but enantiomer 1 cannot
- This implies that the receptors must have a specific shape

2c

c) To calculate the percentage purity of limonene:

- Calculate the number of moles of hydrogen used (with conversion of volume)
  870 $cm^3$ = 0.87 $dm^3$
  **AND**
  Moles of hydrogen = $\frac{0.87}{24.0}$ = 0.03625 moles; [1 mark]
- Calculate the number of moles of limonene that react with the hydrogen
  Moles of limonene = $\frac{0.03625}{2}$ = 0.018125 moles; [1 mark]
- Calculate the mass of limonene used (including $M_r$ calculation)
  Limonene $M_r$ = (12.0 x 10) + (1.0 x 16) = 136 g $mol^{-1}$
  **AND**
  Mass of limonene = 0.018125 x 136 = 2.465 g; [1 mark]
- Calculate the percentage purity

$$\text{Purity} = \frac{\text{mass of pure compound in the impure sample}}{\text{total mass of impure sample}} \times 100 = \frac{2.465}{3.00} \times 100 = 82.2\% \text{ (3sf)};$$

[1 mark]

**[Total: 4 marks]**

- Optical isomers often involve chemical reactions rather than being standalone questions
- This means that an examiner could give you information to complete a variety of calculation questions as part of a larger question containing optical isomers

2d

d) The reagent for this reaction and the type of reaction are:

- Reagent = sodium tetrahydridoborate / sodium borohydride / $NaBH_4$; [1 mark]
- Type of reaction = reduction; [1 mark]

**[Total: 2 marks]**

- Don't be put off by the reaction shown
- If you look at the functional group that changes, you start with a ketone and end with an alcohol
- This is, therefore, reduction and requires sodium tetrahydridoborate

3a

a) Discuss the statement, "aldehydes and ketones always react with HCN to form optical isomers":

Aldehydes

- RCHO + HCN → RC*H(OH)CN; [1 mark]
- If the R-group is a single hydrogen then the product does not show / display / exhibit optical isomerism
  **AND**
  If the R-group is an alkyl group then the product does show / display / exhibit optical isomerism; [1 mark]

Ketones

- R'COR'' + HCN → R'C(CN)(OH)R'' ; [1 mark]
- If the two R-groups are the same then the product does not show / display / exhibit optical isomerism
  **AND**
  If the two R-groups are different then the product does show / display / exhibit optical isomerism; [1 mark]

**[Total: 4 marks]**

- These *discuss* questions are not common as the examiners like to dig a little deeper into various aspects of the reactions of aldehydes and ketones with HCN, such an product nomenclature, reaction mechanisms and 3D representations
- **Tip:** If you're not completely confident with your answer or how to get there, then sketch the molecules out and add the HCN across the carbonyl / C=O bond
- The C=O breaks open and forms a C-O bond
- The hydrogen from HCN adds to the oxygen of the C-O bond
- The cyanide from the HCN adds to the C of the C-O bond
- You don't need to add curly arrows (unless you want to) as this is not outlining the mechanism, you are simply stating what happens

3b

b) To explain the change in chirality, if any, between the β-hydroxyaldehyde and its organic product:

- There is no change in chirality; [1 mark]
- The hydroxyl group is a tertiary alcohol group
  **OR**
  The hydroxyl group is not oxidised by / does not react with acidified potassium $K_2Cr_2O_7$; [1 mark]
- The aldehyde group oxidises to form a carboxylic acid
  **AND**
  The (continued) presence of the carbonyl group means there is no gain in chirality / the compound does not gain a chiral centre; [1 mark]

**[Total: 3 marks]**

- Ignore the language of this question

- A β-hydroxyaldehyde is simply an aldehyde that has a hydroxyl group attached to carbon-3 or the β-position
- This question actually tests your knowledge of the reactions of acidified potassium dichromate(VI) solution
- It also tests your ability to not be distracted by potentially new information or the question itself suggesting that there might be a change in chirality
- The hydroxyl group is a tertiary alcohol and therefore does not oxidise
- The aldehyde will oxidise to form the carboxylic acid as the acidified potassium dichromate(VI) solution is under reflux
- This means that there is no change in chirality of the compound, the chiral centre remains on carbon-3

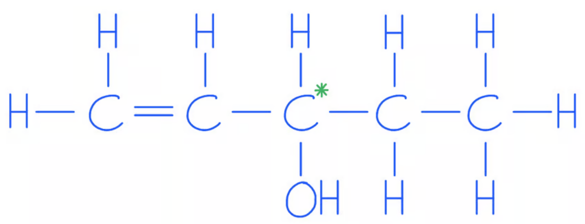
3c

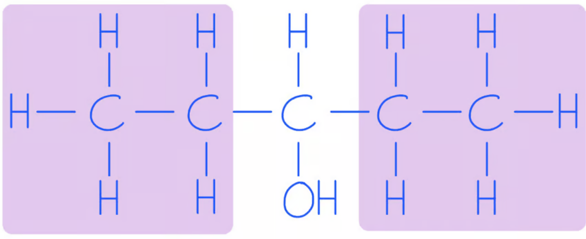
c) The change in chirality, if any, when pent-1-en-3-ol is hydrogenated is:

- There is a loss of chirality; [1 mark]
- (Because) the pent-1-en-3-ol forms pentan-3-ol
  **OR**
  Pent-1-en-3-ol + hydrogen → pentan-3-ol; [1 mark]
- Which forms a symmetrical compound
  **OR**
  Which means that the hydroxyl carbon has 2 ethyl groups / 2 of the same functional groups attached; [1 mark]

**[Total: 3 marks]**

- **Tip:** This is another question where taking the time to sketch the reaction can help secure the marks
- Even just a quick sketch removing the C=C bond and adding a hydrogen atom to each of the carbon atoms, as shown

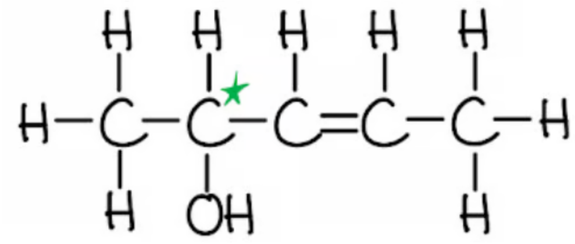




3d

d)

i) The displayed formula of the secondary alcohol isomer of pent-1-en-3-ol that exhibits both stereoisomerism and optical isomerism is:



- Correct position of the C=C bond
  **AND**
  Correct position of the hydroxyl / OH functional group; [1 mark]
- Correct identification of carbon-2 as the chiral centre; [1 mark]

- Correct identification of carbon-2 as the chiral centre; [1 mark]

ii) The IUPAC name for this isomer is:

- Pent-3-en-2-ol; [1 mark]

**[Total: 3 marks]**

- The C=C cannot go on carbon-1 as this would not allow stereoisomerism to occur as one side of the double bond would have 2 hydrogens
- The OH group cannot go on carbon-1 as this would not form a secondary alcohol
- Once you note deduce these points, there are only 3 remaining possibilities
- You can then remove two of these by considering the chirality of the carbons within the chain as shown
- Pent-2-en-4-ol does not get the mark for part (ii)